PHYS4071

Astrophysics & Cosmology

Course Notes

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Introduction

About These Notes

These notes accompany the PHYS4071 Astrophysics & Cosmology course. They are intended as a self-contained supplement to the lectures, providing both conceptual grounding and mathematical rigour at a level appropriate for final-year undergraduates.

The material covers the expanding universe, the Friedmann equations, and the energy budget of the ΛCDM model. Each topic is structured to serve different learning objectives—from building physical intuition to solving examination-style problems and exploring advanced derivations.

These notes do not replace lecture slides or the recommended textbooks. They are best used alongside them as a structured study companion.

Document Conventions

Block Components

Each topic is presented using a consistent **5-component block structure** designed to support multiple learning styles:

PHYSICS INTUITION

Purpose: Builds conceptual understanding through plain language explanations.
This section provides the *motivation* behind the physics concept, explores its *historical development*, and offers *intuitive explanations* that help you grasp what the theory is really about before diving into equations.

Use this when: You're first encountering a new concept or need to refresh the "big picture" before tackling mathematics.

Theory

Purpose: Provides rigorous mathematical treatment of the concept. Here you'll find detailed *derivations*, *fundamental assumptions*, key *equations*, and connections between *mathematics and physical intuition*. This is the technical foundation required for exams and deeper problem-solving.

Use this when: You need to understand how to apply formulas, prepare for exams, or work through problems rigorously.

Question Examples

Purpose: Demonstrates practical application through worked problems.
Each question example mimics exam-style problems and includes complete *solution walkthroughs*. These help you bridge the gap between theory and problem-solving, and expose common pitfalls.

Use this when: Practicing problem-solving, preparing for assessments, or checking if you can apply the formulas correctly.

Foundation

Purpose: Supplies background knowledge and detailed proof steps that underpin the main content.
This section presents *prerequisite mathematics or physics*, *step-by-step derivations* that are too detailed to include in the Theory block, and *intermediate results* needed for a thorough understanding. Reading this is optional for the exam but essential for genuine mastery.

Use this when: A derivation in Theory feels too abrupt, you want to verify an intermediate step, or you need to solidify prerequisite concepts.

Extra

Purpose: Provides supplementary material that extends *beyond* the course syllabus.
This section offers *interesting asides*, *historical context*, *connections to current research*, and citations to deeper references. The content here is intentionally out-of-scope for the course examination but enriches the broader picture of each topic.

Use this when: You're curious to explore beyond the syllabus, or when a topic connects to your personal research interests.

Suggested Reading Paths:

1. **Exam Preparation:** *Physics Intuition* → *Theory* → *Question Examples*
2. **Deep Learning:** *Physics Intuition* → *Foundation* → *Theory* → *Question Examples* → *Extra*
3. **Quick Review:** *Theory* → *Question Examples*

Special Annotations

In addition to the five block components above, two inline annotations appear throughout the text to flag specific types of information:

Important Note

Purpose: Alerts you to a result, condition, or sign convention that is easy to get wrong or is frequently tested.
An Important Note signals a *common misconception*, a *critical sign*, a *non-obvious condition* (e.g. a constraint that must hold for an equation to be valid), or an *exam trap* worth memorising explicitly.

Treat these as mandatory reading regardless of which reading path you follow.

Remark

Purpose: Highlights a useful observation, connection, or naming convention.
A Remark draws attention to a *conceptual link* between results, an *elegant reformulation*, a *naming convention* (e.g. why a quantity carries its particular name), or a *minor but clarifying point* that does not rise to the level of an Important Note.

Use this when: Looking for a deeper understanding of why an equation takes the form it does, or for useful mnemonics.

Mathematical Conventions

The following conventions are used throughout these notes:

Metric signature. We adopt the *mostly-plus* (or "$(-,+,+,+)$") signature:

$$g_{\mu\nu} = \mathrm{diag}(-1,\ +1,\ +1,\ +1).$$

Important Note

Many general-relativity textbooks use the opposite, *mostly-minus* $(+,-,-,-)$ signature. The two conventions differ by an overall sign in the Ricci scalar and

stress-energy trace. Always check the signature convention when comparing formulae across sources.

Natural units. In sections on particle physics or early-universe thermodynamics, we may set $c = \hbar = k_B = 1$. Quantities are then expressed in powers of eV (or GeV). Explicit c and $\hbar$ factors are restored when numerical values are quoted in SI units.

Key symbols.

Symbol	Meaning	Typical units
H_0	Hubble constant today	$\mathrm{km\,s^{-1}\,Mpc^{-1}}$
h	Dimensionless Hubble: $H_0 = 100h$	—
Ω_i	Density parameter for species i	—
$a(t)$	Scale factor ($a_0 = 1$ today)	—
z	Cosmological redshift	—
ρ_{crit}	Critical density	$\mathrm{kg\,m^{-3}}$

Remark

The convention $a_0 \equiv a(t_0) = 1$ is standard and simplifies many formulae, e.g. $1 + z = 1/a(t_e)$. Some older texts normalise a differently; always check the definition when reading the literature.

CHAPTER 1

The Expanding Universe

1.1 Hubble's Discovery

1.1.1 Physics Intuition

PHYSICS INTUITION

In the early twentieth century, astronomers noticed that the spectral lines of distant galaxies were systematically shifted toward the red end of the spectrum. Edwin Hubble's landmark 1929 observations at the Mount Wilson Observatory revealed a clear linear relation between a galaxy's recession velocity and its distance from us. This was the first observational evidence that the universe is not static but expanding—every galaxy is moving away from every other, like raisins in a rising loaf of bread. The discovery overturned centuries of assumptions about a steady, eternal cosmos and laid the empirical foundation for modern Big Bang cosmology.

1.1.2 Theory

THEORY

To quantify Hubble's discovery we infer recession velocity from the Doppler-like redshift of a galaxy's spectral lines: $v = cz$ in the non-relativistic limit. Physical distances are estimated via the *cosmic distance ladder*—at increasing scales: parallax, Cepheid variable stars, and Type Ia supernovae.

The **Hubble–Lemaître Law** relates the recession velocity of a galaxy to its proper distance d:

$$v = H_0 d, \tag{1.1}$$

where $H_0 \approx 67.4\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ is the Hubble constant (Planck 2018). The cosmological redshift is defined via the scale factor $a(t)$:

$$1 + z = \frac{a(t_0)}{a(t_e)} = \frac{1}{a(t_e)}, \tag{1.2}$$

using the convention $a(t_0) = 1$ today.

The Hubble constant H_0 encodes the *current* expansion rate. Direct distance-ladder measurements give $H_0 \approx 73\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, while CMB-based inference (Planck 2018) yields $H_0 \approx 67.4\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$. This $\sim 5\sigma$ discrepancy—the *Hubble tension*—is one of the most pressing open problems in modern cosmology.

1.1.3 Question Examples

Question Example 1.1: Recession Velocity from Redshift

A galaxy in the Virgo Cluster is observed with a spectroscopic redshift of $z = 0.0036$. Using the Hubble–Lemaître Law with $H_0 = 67.4\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, estimate (a) its recession velocity and (b) its distance.

Solution. (a) In the non-relativistic limit, $v = cz = (3 \times 10^5)(0.0036) = 1080\,\mathrm{km\,s^{-1}}$. (b) From $v = H_0 d$, $d = 1080/67.4 \approx 16\,\mathrm{Mpc}$, consistent with the known Virgo Cluster distance of $\sim$ 16–17 Mpc.

1.1.4 Extra

EXTRA

Hubble's original 1929 determination yielded $H_0 \approx 500\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$—nearly eight times the modern value—because he confused two classes of Cepheid variable stars: classical (Population I) Cepheids and the intrinsically dimmer W Virginis stars. Walter Baade's 1952 revision of the Cepheid calibration corrected the distance scale and reconciled the estimated age of the universe with the known ages of the oldest globular clusters.

1.2 The Cosmological Redshift

1.2.1 Physics Intuition

Physics Intuition

The cosmological redshift is fundamentally different from the ordinary Doppler redshift caused by a galaxy's peculiar velocity through space. Instead, it is *space itself* that expands, stretching the wavelength of every photon in transit proportionally to how much the universe has expanded during the photon's journey. The longer a photon travels, the greater its observed redshift—making z a direct measure of cosmic distance and look-back time.

1.2.2 Theory

Theory

Consider a photon emitted at cosmic time t_e with wavelength λ_e, received today (t_0) with wavelength λ_0. In the FLRW metric, the null geodesic condition $ds^2 = 0$ for radial travel implies that the proper wavelength scales with the scale factor: $\lambda \propto a(t)$. Therefore

$$\frac{\lambda_0}{\lambda_e} = \frac{a(t_0)}{a(t_e)} = 1 + z, \tag{1.3}$$

reproducing Eq. (1.2). This derivation requires only the FLRW metric—no specific model for $a(t)$ is needed. In the limit of small separations, $z \ll 1$, it reduces to the familiar Doppler formula $z \approx v/c$ with $v = H_0 d$.

1.2.3 Question Examples

Question Example 1.2: Computing Cosmic Distance

A hydrogen Lyman-α photon (rest wavelength $\lambda_e = 121.6\,\text{nm}$) from a distant quasar is detected at $\lambda_0 = 729.6\,\text{nm}$. Find the redshift z and the scale factor $a(t_e)$ at the time of emission.

Solution. $z = \lambda_0/\lambda_e - 1 = 729.6/121.6 - 1 = 5.0$. From $1 + z = 1/a(t_e)$, we get $a(t_e) = 1/6 \approx 0.167$: the universe was six times smaller when this photon was emitted.

1.2.4 Extra

Extra

The most distant objects confirmed to date reach redshifts $z > 13$, placing them within the first few hundred million years after the Big Bang. JWST has revealed surprisingly luminous, massive galaxies at $z > 10$, challenging standard semi-analytic models of early galaxy formation. Such observations probe the *epoch of reionization*—when ultraviolet radiation from the first stars and quasars ionized the neutral hydrogen permeating the universe.

CHAPTER 2

Friedmann Equations & Cosmological Dynamics

2.1 The FLRW Metric

2.1.1 Physics Intuition

PHYSICS INTUITION

The principle of *cosmic homogeneity and isotropy*—that on large scales the universe looks the same from every point and in every direction—drastically simplifies the mathematical description of spacetime. This is not merely a convenient assumption: CMB temperature maps show variations of only one part in 10^5, and galaxy surveys above $\sim 100\,\mathrm{Mpc}$ are remarkably uniform. These two symmetries together constrain the spacetime metric to the unique Friedmann–Lemaître–Robertson–Walker (FLRW) form.

2.1.2 Theory

THEORY

The **FLRW metric** describes a homogeneous, isotropic universe:

$$ds^2 = -c^2\,dt^2 + a(t)^2\left[\frac{dr^2}{1-kr^2} + r^2\,d\theta^2 + r^2\sin^2\theta\,d\phi^2\right], \tag{2.1}$$

with curvature parameter $k \in \{-1, 0, +1\}$.
The curvature parameter k determines the global geometry of spatial slices: $k = +1$ for a closed (sphere-like) universe, $k = -1$ for an open (saddle-like) universe, and $k = 0$ for a spatially flat (Euclidean) universe. The comoving coordinate r labels a fixed point in the matter distribution; physical (proper) distances scale with the scale factor as $d_{\mathrm{phys}} = a(t)\,r$. Current Planck observations constrain $|\Omega_k| < 0.005$, consistent with exact flatness.

2.1.3 Question Examples

Question Example 2.1: Proper Distance in FLRW

Two galaxies are separated by a comoving distance $r = 1000\,\text{Mpc}$ in a flat ($k = 0$) FLRW universe. Find their proper separation (a) today and (b) at $z = 2$, when $a(t_e) = 1/3$.

Solution. (a) Today $a(t_0) = 1$, so $d_{\text{phys}} = 1000\,\text{Mpc}$. (b) At $z = 2$, $d_{\text{phys}} = a(t_e) \cdot r = (1/3) \times 1000 \approx 333\,\text{Mpc}$; the universe was three times smaller and the galaxies three times closer together.

2.1.4 Extra

EXTRA

The FLRW metric was independently derived by Alexander Friedmann (1922), Georges Lemaître (1927), Howard Robertson (1935), and Arthur Walker (1936). Lemaître was the first to combine it with recession-velocity data to argue that the universe is expanding—two years before Hubble's 1929 paper—and also extrapolated backwards to propose what he called the *primeval atom*, now known as the Big Bang.

2.2 The Friedmann Equations

2.2.1 Physics Intuition

PHYSICS INTUITION

The Friedmann equations are the equations of motion for the universe as a whole. The first equation resembles Newtonian energy conservation: the square of the expansion rate ($H = \dot{a}/a$) equals the total energy density divided by an appropriate factor. The second (acceleration) equation tells us whether expansion speeds up or slows down: ordinary matter and radiation have positive pressure and *decelerate* expansion, while a cosmological constant (dark energy) with $w = -1$ *accelerates* it.

2.2.2 Theory

THEORY

Inserting the FLRW metric into the Einstein field equations gives the **Friedmann**

equations:

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho - \frac{kc^2}{a^2} + \frac{\Lambda c^2}{3}, \tag{2.2}$$

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\left(\rho + \frac{3p}{c^2}\right) + \frac{\Lambda c^2}{3}. \tag{2.3}$$

The equation-of-state parameter $w = p/(\rho c^2)$ characterises each component: $w = 0$ for pressureless dust (cold dark matter or baryons), $w = 1/3$ for radiation and relativistic species, and $w = -1$ for a cosmological constant. Components with $w < -1/3$ contribute positively to the acceleration Eq. (2.3), driving accelerated expansion. Combining both Friedmann equations yields the *fluid equation*, $\dot{\rho} + 3H(\rho + p/c^2) = 0$, expressing covariant energy conservation for each component independently.

2.2.3 Question Examples

Question Example 2.2: Scale Factor of a Flat Matter-Dominated Universe

Derive the time dependence $a(t)$ for a flat, radiation-dominated universe ($k = 0$, $\Lambda = 0$, $w = 1/3$).

Solution. For $k = 0$, $\Lambda = 0$ and pressureless dust ($p = 0$), Eq. (2.2) reduces to $\dot{a}^2 \propto a^{-1}$, giving:

$$a(t) \propto t^{2/3}. \tag{2.4}$$

For radiation ($w = 1/3$), $\rho \propto a^{-4}$, so $\dot{a}^2 \propto a^{-2}$ and $a \propto t^{1/2}$—a slower expansion than the matter-dominated case.

2.2.4 Extra

EXTRA

Friedmann published his expanding-universe solution in 1922; Einstein initially published a rebuttal claiming a mathematical error, then retracted it. It was not until Hubble's 1929 observational evidence that the scientific community accepted the expanding universe. The steady-state theory—championed by Hoyle, Gold, and Bondi until the 1960s—proposed continuous matter creation to maintain constant mean density, but was decisively ruled out by the discovery of the cosmic microwave background in 1965.

CHAPTER 3

Energy Budget in the ΛCDM Model

3.1 Density Parameters & Critical Density

3.1.1 Physics Intuition

PHYSICS INTUITION

The *critical density* ρ_{crit} is the threshold total energy density at which the universe has exactly flat geometry. A universe denser than ρ_{crit} would be positively curved and—in the absence of dark energy—would re-collapse in a Big Crunch; one less dense would be negatively curved and expand forever. Observations show our universe is within 0.5% of the critical density, implying either extraordinary initial fine-tuning or a dynamical mechanism such as cosmological inflation that drove it toward flatness.

3.1.2 Theory

THEORY

Define the **critical density** and dimensionless density parameters:

$$\rho_{\text{crit}}(z) = \frac{3H(z)^2}{8\pi G}, \qquad \Omega_i(z) = \frac{\rho_i(z)}{\rho_{\text{crit}}(z)}. \tag{3.1}$$

Each component scales with redshift according to its equation of state w_i:

$$\rho_i(z) = \rho_{i,0}(1+z)^{3(1+w_i)}. \tag{3.2}$$

The flatness condition $\sum_i \Omega_i = 1$ (for $k = 0$) is equivalent to $\Omega_{\text{m},0} + \Omega_\Lambda + \Omega_{\text{r},0} = 1$ in a universe containing only matter, radiation, and a cosmological constant. A non-zero spatial curvature contributes a term $\Omega_k = -kc^2/(a_0 H_0)^2$ to the sum, shifting it away from unity.

IMPORTANT NOTE

The density parameter Ω_k for curvature is *not* a physical energy component; it is a bookkeeping term that ensures the Friedmann equation takes the compact form $\sum_i \Omega_i + \Omega_k = 1$. Do not confuse it with matter or radiation density: $\Omega_k < 0$ corresponds to positive curvature ($k = +1$), and $\Omega_k > 0$ to negative

curvature ($k = -1$).

FOUNDATION

Derivation of ρ_{crit} from the Friedmann equation. Recall the first Friedmann equation (for a flat or curved universe):

$$H^2 = \left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho_{\text{tot}} - \frac{kc^2}{a^2}.$$

Setting the spatial curvature to zero ($k = 0$) defines the critical total energy density required to achieve exact flatness:

$$H^2 = \frac{8\pi G}{3}\rho_{\text{crit}} \implies \rho_{\text{crit}} = \frac{3H^2}{8\pi G}.$$

For an arbitrary universe with $\rho_{\text{tot}} \neq \rho_{\text{crit}}$, dividing both sides of the Friedmann equation by H^2 gives

$$1 = \frac{\rho_{\text{tot}}}{\rho_{\text{crit}}} - \frac{kc^2}{a^2H^2} \equiv \Omega_{\text{tot}} + \Omega_k,$$

showing that $\Omega_k \equiv -kc^2/(aH)^2$ exactly accounts for the departure from flatness.

REMARK

The word "critical" refers to the fact that ρ_{crit} is the *threshold* between an open and a closed universe in the absence of a cosmological constant. In modern ΛCDM cosmology this geometric interpretation is supplemented by the role dark energy plays: even with $\Omega_k \approx 0$, the universe accelerates rather than re-collapsing.

3.1.3 Question Examples

Question Example 3.1: Computing Critical Density Today

Calculate ρ_{crit} at $z = 0$ using $H_0 = 67.4\,\text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$. Express your answer in $\text{kg}\,\text{m}^{-3}$ and in protons per cubic metre.

Solution. Convert to SI: $H_0 = 67400/(3.086 \times 10^{22})\,\text{s}^{-1} = 2.18 \times 10^{-18}\,\text{s}^{-1}$. Then $\rho_{\text{crit}} = 3H_0^2/(8\pi G) \approx 8.5 \times 10^{-27}\,\text{kg}\,\text{m}^{-3}$. Dividing by $m_p = 1.67 \times 10^{-27}$ kg gives ≈ 5.1 protons per cubic metre—an astonishingly sparse universe.

3.1.4 Extra

Extra

The flatness of the universe is closely tied to inflationary cosmology: a brief period of exponential expansion at $t \sim 10^{-36}$ s stretches any initial curvature to imperceptibly small values, predicting $|\Omega_k| \ll 1$ today. This prediction was confirmed to $< 0.5\%$ by the positions of acoustic peaks in the Planck CMB power spectrum—one of inflation's most compelling successes.

3.2 The Flat ΛCDM Model

3.2.1 Physics Intuition

Physics Intuition

The ΛCDM model is the standard model of cosmology, successfully describing observations from the CMB to the large-scale structure of galaxies. Its dominant components are: baryonic matter ($\sim 5\%$), cold dark matter ($\sim 27\%$), and a cosmological constant Λ (dark energy, $\sim 68\%$). Despite its observational success, the physical nature of both dark matter and dark energy remain unknown—arguably the two greatest open questions in fundamental physics today.

3.2.2 Theory

THEORY

In the **flat ΛCDM** model ($\Omega_k = 0$, $\Omega_{\mathrm{m},0} + \Omega_\Lambda + \Omega_{\mathrm{r},0} = 1$):

$$\frac{H(z)^2}{H_0^2} = \Omega_{\mathrm{m},0}(1+z)^3 + \Omega_{\mathrm{r},0}(1+z)^4 + \Omega_\Lambda. \tag{3.3}$$

Current Planck 2018 best-fit values:

$$\Omega_{\mathrm{m},0} \approx 0.315, \quad \Omega_\Lambda \approx 0.685, \quad \Omega_{\mathrm{r},0} \approx 9.2 \times 10^{-5}.$$

The cosmological constant Λ (or equivalently, dark energy with equation of state $w_\Lambda = -1$) constitutes roughly two-thirds of the universe's energy budget today. Its value and physical interpretation remain deeply mysterious [1].
The energy budget evolves dramatically with redshift. At early times radiation dominated ($\rho_\mathrm{r} \propto a^{-4}$ dilutes fastest); the *matter–radiation equality* at $z_\mathrm{eq} \approx 3400$ marks the transition to the matter-dominated era that shaped large-scale structure. Dark energy overtook matter around $z \approx 0.4$, initiating the current phase of accelerated expansion. As $z \to -1$ (far future), $H(z) \to H_0\sqrt{\Omega_\Lambda}$—the universe approaches an exponential de Sitter expansion.

3.2.3 Question Examples

Question Example 3.2: Age of the Universe in Flat ΛCDM

Using Planck 2018 parameters, evaluate the age integral $t_0 = \int_0^\infty dz/[(1+z)H(z)]$ and show that it gives $t_0 \approx 13.8\,\mathrm{Gyr}$.

Solution. The lookback-time integral is:

$$t_0 = \int_0^\infty \frac{dz}{(1+z)\,H(z)}, \tag{3.4}$$

with $H(z)$ from Eq. (3.3). Substituting Planck 2018 values and numerically integrating (e.g. with `scipy.integrate.quad`) yields $t_0 \approx 13.8\,\mathrm{Gyr}$, consistent with the ages of the oldest globular clusters (12–13 Gyr).

3.2.4 Extra

Extra

The *Hubble tension*—the $\sim 5\sigma$ discrepancy between early-universe (CMB-based, $H_0 \approx 67.4$) and late-universe (Cepheid/SN Ia distance ladder, $H_0 \approx 73$) measurements of H_0 in units of $\mathrm{km\,s^{-1}\,Mpc^{-1}}$—remains unresolved. Proposed extensions to ΛCDM include early dark energy, self-interacting dark matter, or extra relativistic species. Whether this tension signals new physics or unaccounted systematics is one of the most actively debated questions in observational cosmology.

3.3 Energy Budget Evolution

3.3.1 Visualizing the Energy Density

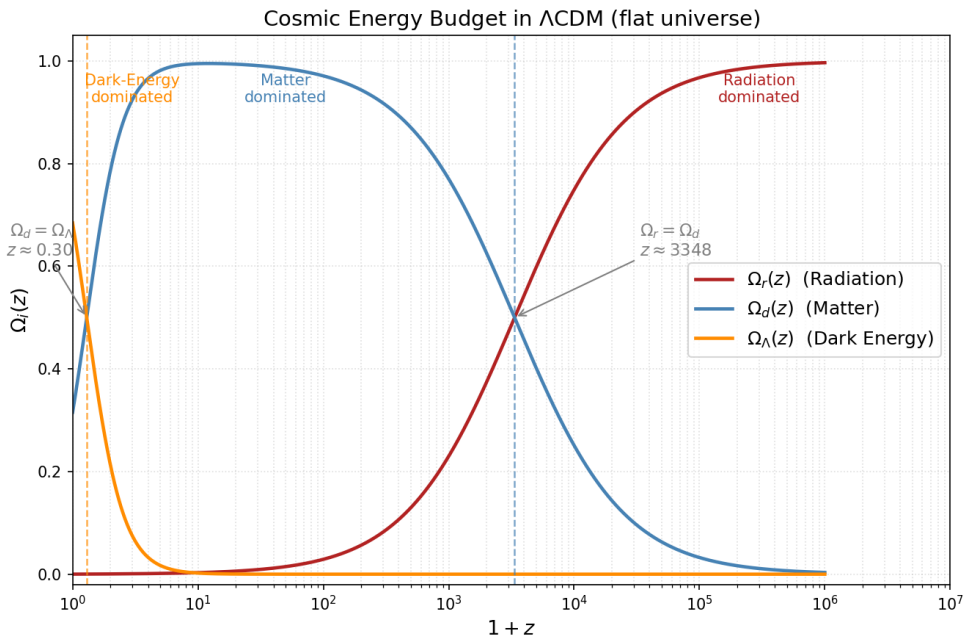

Figure 3.1: **Evolution of energy density components in the flat ΛCDM model.** The figure shows how the fractional energy density (normalized to critical density) of matter (dark and baryonic), radiation, and dark energy evolve with redshift z. At early times ($z \gg 1$), radiation dominated; at intermediate times, matter dominated; and at late times ($z \lesssim 1$), dark energy becomes dominant.

Extra

The visualization of energy budget components provides crucial insights into cosmic evolution. Early universe physics was radiation-dominated, while the matter-dominated era shaped structure formation and galaxy evolution. The recent transition to dark-energy domination (around $z \sim 0.7$) explains the observed cosmic acceleration detected through Type Ia supernovae observations [1].

CHAPTER A

Declaration of AI Assistance

A.1 Use of Artificial Intelligence in This Document

This course guide was developed with assistance from artificial intelligence (AI) in the following capacities:

- **Template Design:** AI provided guidance on structuring the document layout, choosing appropriate color schemes.
- **LaTeX Code Generation:** AI assisted in writing and refactoring LaTeX code, including package configurations, box styling (tcolorbox), TikZ decorations, and custom command definitions.
- **LaTeX Structure Refactoring:** AI helped reorganize and optimize the document structure for maintainability, readability, and professional presentation.
- **Mathematical Formula Formatting:** AI provided assistance in formatting and presenting mathematical equations in clear, readable notation aligned with conventions.

Author Verification: All content, including scientific accuracy, theoretical derivations, example problems, and pedagogical choices, has been carefully reviewed and verified by the author. The AI assistance was limited to technical formatting and structural optimization, not to the generation or validation of course content itself.

Compiled by: automatoc-ti | *Date: Sunday 8th March, 2026*

CHAPTER B

Mathematical Tools & Derivations

B.1 Useful Identities

Trigonometric

$$\sin^2\theta + \cos^2\theta = 1, \quad \tan\theta = \sin\theta/\cos\theta, \quad \sin(A \pm B) = \sin A\cos B \pm \cos A \sin B.$$

Hyperbolic

$$\cosh^2 x - \sinh^2 x = 1, \quad \sinh(x \pm y) = \sinh x \cosh y \pm \cosh x \sinh y.$$

Common Integrals

$$\int x^n\,dx = \tfrac{x^{n+1}}{n+1} + C \quad (n \neq -1), \qquad \int e^{ax}\,dx = \tfrac{1}{a}e^{ax} + C,$$

$$\int \frac{dx}{x} = \ln|x| + C, \qquad \int \sin x\,dx = -\cos x + C.$$

B.2 Differential Geometry Primer

The **metric tensor** $g_{\mu\nu}$ encodes the geometry of spacetime: $ds^2 = g_{\mu\nu}\,dx^\mu\,dx^\nu$. The **Christoffel symbols** (not tensors) are defined as

$$\Gamma^\lambda_{\mu\nu} = \tfrac{1}{2}g^{\lambda\sigma}(\partial_\mu g_{\nu\sigma} + \partial_\nu g_{\mu\sigma} - \partial_\sigma g_{\mu\nu}). \tag{B.1}$$

The **Ricci tensor** is the contraction $R_{\mu\nu} = R^\lambda{}_{\mu\lambda\nu}$ of the Riemann curvature tensor, and the Ricci scalar is $R = g^{\mu\nu}R_{\mu\nu}$. These enter the Einstein field equations as $G_{\mu\nu} \equiv R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = (8\pi G/c^4)T_{\mu\nu}$.

CHAPTER C

Constants & Conversion Factors

C.1 Physical Constants

Quantity	Symbol	Value (SI)
Speed of light	c	$2.998 \times 10^{8}\,\mathrm{m\,s^{-1}}$
Gravitational constant	G	$6.674 \times 10^{-11}\,\mathrm{m^{3}\,kg^{-1}\,s^{-2}}$
Hubble constant today	H_0	$67.4 \pm 0.5\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$

C.2 Conversion Factors

Conversion	Value
1 parsec (pc)	$3.086 \times 10^{16}\,\mathrm{m}$
1 kiloparsec (kpc)	$3.086 \times 10^{19}\,\mathrm{m}$
1 megaparsec (Mpc)	$3.086 \times 10^{22}\,\mathrm{m}$
1 gigaparsec (Gpc)	$3.086 \times 10^{25}\,\mathrm{m}$
1 light-year (ly)	$9.461 \times 10^{15}\,\mathrm{m}$
1 gigalight-year (Gly)	$9.461 \times 10^{24}\,\mathrm{m}$
1 year (yr)	$3.156 \times 10^{7}\,\mathrm{s}$
1 Gyr	$3.156 \times 10^{16}\,\mathrm{s}$

REFERENCES

[1] Moshe Carmeli. "Value of the cosmological constant: Theory versus experiment". In: *AIP Conference Proceedings*. Vol. 586. AIP, 2001, pp. 316–318. DOI: 10.1063/1.1419571. URL: http://dx.doi.org/10.1063/1.1419571.